Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/06/2013
PI Name:	Nielsen, Sheila Ph.D.		
Project Title:	Genotypic and phenotypic responses of Candida albicans to spaceflight		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
PI Email:	sheila.nielsen@montana.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	406-994-5177
Organization Name:	Montana State University		
PI Address 1:	Leon Johnson Hall, room 315		
PI Address 2:			
PI Web Page:			
City:	Bozeman	**State:**	MT
Zip Code:	59717	**Congressional District:**	1
Comments:	PI name change to Sheila Nielsen in 2014 (formerly Sheila Nielsen-Preiss)--Ed., 1/12/2015		
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	04/01/2012	**End Date:**	08/31/2015
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
Contact Email:	jeffrey.d.smith2@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	Space-X ; ISS NOTE: Extended to 8/31/2015 for reflight as Micro-8 experiment, per A. Chu/ARC and NSSC information (Ed., 4/3/14) NOTE: Extended to 8/31/2014 per A. Chu/ARC and NSSC information (Ed., 6/17/13)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX12AH29G		
Performance Goal No.:			

Performance Goal Text:

Task Description:

Micro-6 Experiment
(Addendum to: "The fungal response to simulated microgravity, grant NNX09AM79G")

Modeling the microgravity environment, using rotating suspension culture bioreactors, has allowed investigation of eukaryotic cellular responses in a ground-based model and has provided insights into the impact of spaceflight on cellular physiology. Extrapolation of results from modeled systems to that expected in space is useful and informative, but evaluating the impact of actual space exposure strengthens the validity and interpretation of data from both environments. Using the Candida albicans (C. albicans) yeast system, we have demonstrated that simulated microgravity affects differential expression of several groups of genes, including those related to morphogenic transformation, environmental sensing and ergosterol biosynthesis; induces morphologic changes such as filamentation, biofilm formation and colony morphology; and results in an increased resistance to the antimicrobial agent Amphotericin B. The constellation of these physiological responses is consistent with a potential increase in the virulence of this opportunistic yeast pathogen. The focus of the current proposal is to characterize physiological outcome measures of C. albicans following exposure to flight on the International Space Station (ISS). The overall objective of this proposal is to evaluate the response of C. albicans to spaceflight in order to 1) predict the potential health risk posed by this opportunistic pathogen in the spaceflight environment, and 2) to further characterize the phenotypic and genotypic response of a eukaryotic system to conditions of microgravity.

Micro-8 Experiment (Addendum March 2014 to "Genotypic and phenotypic responses of Candida albicans to spaceflight")

We had the opportunity to fly the Micro-6 payload of C. albicans experiments on SpX CRS-1 in October 2012 and many of the experimental results from this study were positive and intriguing. Subsequent to this initial flight experience, we are in a position to repeat several of the experiments to validate the data and provide power to the analyses. In addition, pursuant to the lessons learned, internally by our team and more generally as part of the initial Space Exploration expedition, we feel refining the experimental design and altering the sample handling processes will strengthen the data.

The goal of our ongoing studies is to further examine the responses of C. albicans to microgravity by performing hypothesis-driven research in spaceflight conditions. Our overarching hypothesis is that exposure of C. albicans to microgravity will alter gene expression and morphology, consistent with a potential increase in virulence. Notably, these studies 1) will inform the value of LSMMG for predicting the physiological responses of C. albicans, 2) will further explore and document the phenotypic parameters of C. albicans that are associated with pathogenicity and are altered during exposure to spaceflight, and 3) may predict conserved responses of higher eukaryotes, including humans, to spaceflight conditions. This re-flight opportunity of the C. albicans payload will strengthen and extend the data as well as focus the analysis and interpretation.

(Ed. note 4/8/14: reporting for the Micro-8 addendum will begin in July 2014)

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The microgravity environment is also one of low fluid shear. Some of our unpublished data suggests that this low fluid shear environment may be contributing significantly to the yeast response. Although there are clear implications for yeast grown in space, there are also microenvironments within the body that might be classified as low fluid shear, such as the brush border regions in the digestive tract. We hope to be able to extrapolate the results from the low shear flight environment to the response of Candida albicans in the human digestive system.

Task Progress:

The goals of our current studies were to evaluate the effects of spaceflight on the behavior and characteristics of the opportunistic yeast pathogen, Candida albicans. Our previous studies using modeled microgravity (LSMMG, low shear modeled microgravity) suggested that the yeast adapted to the environment of low shear by assuming characteristics consistent with increased aggressiveness. We were fortunate to fly a set of experiments as the Micro-6 payload on the initial Space Exploration Cargo Resupply Service flight in October 2012.
Initial yeast samples were prepared in our home laboratory in Montana and flown to Kennedy Space Center (KSC) for preparation of flight experiments. The research team consisted of the PI and 5 students, one of whom has recently left the laboratory to begin medical school. Samples were prepared at KSC with the assistance of the NASA support staff and BioServe and submitted to SpaceX for transport to the International Space Station (ISS) on October 8, 2012. Ultimately samples were processed at different times over the course of the 17 days on the ISS. Samples were returned via Dragon on October 28, 2012 and retrieved at Long Beach, California. All samples were returned to Montana for analysis. Many samples were fixed on the ISS so the transport temperature was a bit flexible. Unfortunately, the viable samples were exposed to growth-inducing temperatures (ambient) and not analyzed for up to 4 days so space-induced adaptations were impossible to evaluate. Ideally and in the future, viable samples should be returned in chilled temperatures.

Growth rates in the microgravity environment appear to be organism specific; therefore, one of our baseline inquiries was to determine how microgravity affected the growth of C. albicans. Growth was stimulated by the addition of medium once on the ISS and, following a defined incubation time, cell growth was halted by fixing the samples. There was not a statistically significant difference in the growth of samples grown on the ISS as compared to ground-control matched samples. But interestingly, there was an indication that samples exposed to the microgravity environment for longer periods of time might experience a slightly diminished growth capacity. If a reflight opportunity becomes available, yeast will be exposed to the microgravity environment for even longer periods of time prior to growth arrest and fixation.

Ground-based studies demonstrated that C. albicans exposed to the low shear environment became more resistant to the antifungal agent Amphotericin B. Naturally, we were very interested to evaluate whether this response was also true of samples grown in space. Indeed, yeast grown on the ISS were significantly more resistant to high concentrations of the antifungal agent than their ground-based controls.

Finally, many of the adaptation responses we queried are simply predictors of virulence. A direct analysis of virulence is the gold standard and there are many models for determining yeast virulence. In order to get a glimpse of the aggressiveness of C. albicans following exposure to microgravity, viable samples were used to infect human monocytes as the host model. Interestingly, monocytes were less able to resist the challenge posed by yeast grown in space as compared to those grown in the ground-based cultures. We are currently evaluating the monocyte cytokine response with the hope of testing frozen samples obtained immediately following the flight experiment.

	It is worth noting that not all the phenotypic responses were as we predicted. We had observed a dramatic transition in colony morphology in cells grown in LSMMG that was not replicated from the cells cultured on the ISS. We are also still investigating the gene expression profile and the biofilm production of cells grown in space relative to their ground-based counterparts. This flight opportunity was our first and presented a unique opportunity to analyze C. albicans' responsiveness to spaceflight. We were able to obtain data from nearly all of the experiments and have significant analysis yet to do. We also learned enough from these experiments to design targeted repeat analyses, as well as to develop appropriate follow-up studies should a re-flight come available. Evaluation of the virulence capacity of C. albicans in adaptation to extreme environments is important for several reasons. Analysis of the immune cells from flight crews suggest that there is a diminished immune response during space travel; therefore, increased virulence on the part of microbes could pose a health risk. In addition, the low shear environment of microgravity may resemble niches within the human body, such as the brush border regions within the gastrointestinal tract, so an understanding of the yeast response may be extrapolated to disease control and treatment on Earth.
Bibliography Type:	Description: (Last Updated: 06/23/2023)
Abstracts for Journals and Proceedings	Sutton J. "Effects of Microgravity on Candida-induced Cytokine Response of Human Monocytes." Hughes Undergraduate Biology Complex Biological Systems Summer Research Program, Montana State University, August 6, 2013. Oral and Poster Presentations given by James Sutton, MSU undergraduate student. Hughes Undergraduate Biology Complex Biological Systems Summer Research Program, Montana State University, August 6, 2013. , Aug-2013
Abstracts for Journals and Proceedings	Nielsen-Preiss S. "Evaluation of the Response of C. albicans to Spaceflight." 2nd Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 16-18, 2013. 2nd Annual International Space Station (ISS) Research and Development Conference, Denver, CO, June 16-18, 2013. , Jun-2013
Significant Media Coverage	Cantrell A. "Article in Montana State U's campus newsletter about the PI's flight experiment: 'MSU team to send experiments to International Space Station.' " Montana State University's campus newsletter, Sept 17, 2012. http://www.montana.edu/news/11461/msu-team-to-send-experiments-to-international-space-station ; accessed 8/8/2013., Sep-2012
Significant Media Coverage	Cantrell A. "Article about PI's flight experiment: 'MSU team to send experiments to International Space Station.' " Bozeman Daily Chronicle. September 24, 2012. http://www.bozemandailychronicle.com/news/montana_state_university/article_2db1b990-05fc-11e2-9181-0019bb2963f4.html ; accessed 8/8/2013., Sep-2012
Significant Media Coverage	Patterson B, McMillan S. "Online article about PI's flight experiment. 'Yeast Studies in Microgravity Raising New Treatment Potential For Infections on Earth.' " ISS online news/NASA Ames Research Center, 9/28/2012. http://www.nasa.gov/mission_pages/station/research/news/micro_6.html ; accessed 8/8/2013., Sep-2012
Significant Media Coverage	Maschmedt L. "Story and video about PI's flight experiment: 'MSU study to launch into space.' " NBC Montana, station KTVM. Story and video by Lauren Maschmedt, Sept. 18, 2012., Sep-2012
Significant Media Coverage	Slate J. "News segment on PI's flight experiment: 'MSU experiment heading to outer space.' " Montana CBS News segment, 9/20/12., Sep-2012
Significant Media Coverage	NASA-TV news briefing with Julie Robinson. "NASA-TV news briefing with Julie Robinson, including PI's experiment." NASA-TV news briefing with Julie Robinson, including PI's experiment, October 7, 2012., Oct-2012
Significant Media Coverage	Orlando CBS. "Orlando CBS news coverage: SpaceX launches for the International Space Station." Orlando CBS (Oct 7, 2012) http://www.clickorlando.com/news/SpaceX-launches-for-the-International-Space-Station/-/1637132/16893284/-/9ari30/-/index.html ; accessed 8/8/2013., Oct-2012